

The Development of WLAN Time Face Recorder System

Sirilux Keawoud¹, Rapheephon Riyapraw¹, Siriyakorn Yuyen¹, Suppakarn Chansareewittaya^{1,2},

and Mahamah Sebakor^{1,2,*}

¹School of Information Technology, Mae Fah Luang University

²Computer and Communication Engineering for Capacity Building Research Center, Mae Fah Luang University

*Corresponding author: mahamah.seb@mfu.ac.th

Abstract—Nowadays, we have various time recording systems. One of them is made by face detection whose aim is to make the system more convenient. This work presents the alternate application system called the WLAN time face recorded system which uses a wifi network and face detection for a time attendance system. Face detection by OpenCV library. We take advantage of WLAN technology and personal devices such as mobile phones to verify user attendance in their organization. This work is significantly useful and convenient for the organization to increase the reliability of attendance verification.

Keywords: WLAN, mac address, face detection, pattern recognition; recorder system

I. INTRODUCTION

Check attendance is what organizations use to track employee attendance and is also used as one of the variables for evaluating work performance. Normally, most companies prefer to use fingerprint scanners or face recognition attendance system. Some companies may install too few of these machines. So, we created a WLAN time face recorder system that is pilot, as a guideline for solving the problem. By implementing this technology, organizations can streamline their attendance tracking process, reduce wait times, and ensure more accurate and fair evaluations of employee work performance. Additionally, it aligns with the trends in modern workforce management and provides employees with a convenient tool for managing their attendance.

The use of WLAN, WLC, and face detection in the authentication process of a check attendance system can indeed provide several benefits and help prevent errors and fraudulent activities. Here's how these technologies contribute to the identity verification process: (1) WLAN (Wireless Local Area Network): WLAN enables wireless communication and data transmission using radio waves or infrared waves. By utilizing WLAN, the attendance system can provide flexibility in connectivity. Employees can connect to the network using their mobile devices, such as smartphones, laptops, or tablets. This eliminates the need for physical connections and allows users to access the attendance system from anywhere within the network range. (2) WLC (Wireless LAN Controller): WLC acts as a central controller for all Access Points (APs) in a WLAN network. APs receive the internet signal from the service provider and broadcast it to extend the range of WLAN

connections. The use of WLC ensures seamless connectivity and consistent signal strength throughout the network. It helps maintain a stable connection for users, ensuring reliable authentication and attendance tracking. (3) MAC Address: MAC address is a unique identifier assigned to network devices, such as computers, routers, or smartphones. In the attendance system, MAC addresses can be used to verify the identity of the devices connecting to the network. By associating specific MAC addresses with employee profiles, the system can ensure that only authorized devices are used for attendance checking, minimizing the possibility of fraudulent activities. (4) Face Detection and Recognition: Face detection is the process of locating and identifying faces within images or videos. In the attendance system, face detection can be employed to capture the employee's face during the check-in/check-out process. The system can compare the captured face with a pre-existing database of authorized faces to perform face recognition. This helps prevent cheating attempts, such as someone else checking attendance on behalf of another employee.

By combining these technologies, the attendance system can provide a convenient and reliable method for employees to verify their identities. The system can leverage WLAN for wireless connectivity, utilize WLC for seamless network coverage, validate MAC addresses for device authentication, and employ face detection and recognition for accurate identification. These measures enhance the overall security and integrity of the attendance system, making it more resistant to errors and fraudulent activities.

II. RELATED WORK

Various technologies have been used in this paper. This section briefly introduces some related work of our design architecture, including WLAN, face detection, MAC address, and WLC enabling our design to succeed. We will start with the description of related work on the network. In [1], this paper introduces the importance of face detection in this digital age, and what it can do, including advantages and limitations. In this topic [2], they will discuss MAC address verification, and MAC address reservation according to actual users. Even recording the MAC address from the user's computer [3] will explain the factors of communication by using WLAN. This paper will provide an in-depth study on WLAN and related issues [4] AttenFace system has the potential to offer a convenient

solution for attendance tracking in educational institutions, it's important to carefully consider the technical, ethical, privacy, and operational aspects of its implementation. Additionally, engaging with stakeholders and obtaining feedback during the development and deployment phases will be critical to its success. [5] the proposed automated attendance system with implicit authentication using Wi-Fi and mobile devices has the potential to streamline attendance recording. However, it's important to address technical challenges, privacy concerns, and user acceptance to ensure its successful implementation in educational or organizational settings. Additionally, conducting thorough testing and assessing the practicality of the system is essential before widespread adoption. [6] AASs have the potential to bring efficiency and accuracy to attendance tracking in educational institutions, their successful implementation requires careful planning, consideration of privacy and security concerns, and proper integration with existing systems. When implemented and utilized effectively, AASs can be a valuable tool in maintaining accurate attendance records. [7] the proposed face recognition system for mobile wireless devices shows promise in terms of speed and improved battery life. However, it's important to address issues related to network connectivity, data security, and user experience to ensure its successful implementation and adoption in real-world applications. Additionally, considering scalability and compatibility with various devices can contribute to its broader usability.

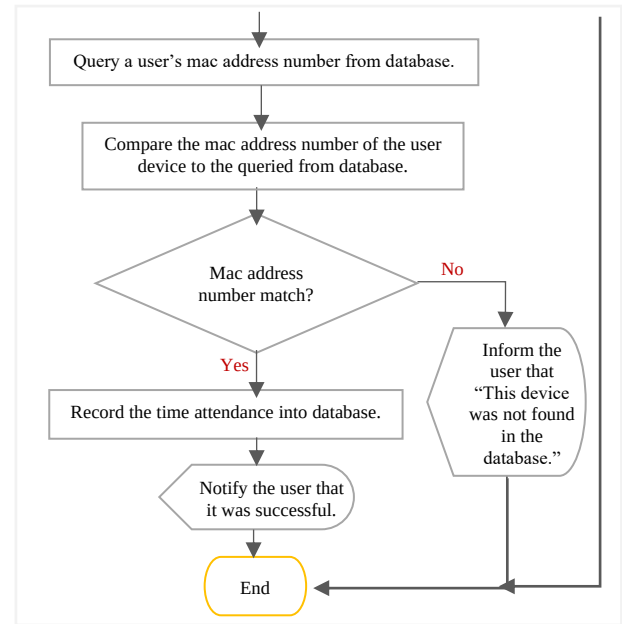


Fig. 1. System flowchart

I. PROPOSED APPROACH

To make the time face recorder system more convenient to use by adopting WLAN for the authentication part, the time face recorder system can be used with the user's personal device. System flowchart is shown in Fig. 1.

The OpenCV is used for face detection [8-10]. The database will store data according to ER diagram as in Fig. 2. and collect 4 photos of the user, including a face from a distance, a face upright, a face straight while smiling, a face with eyes closed, as in Fig. 3. System overview is described as in Fig. 4.

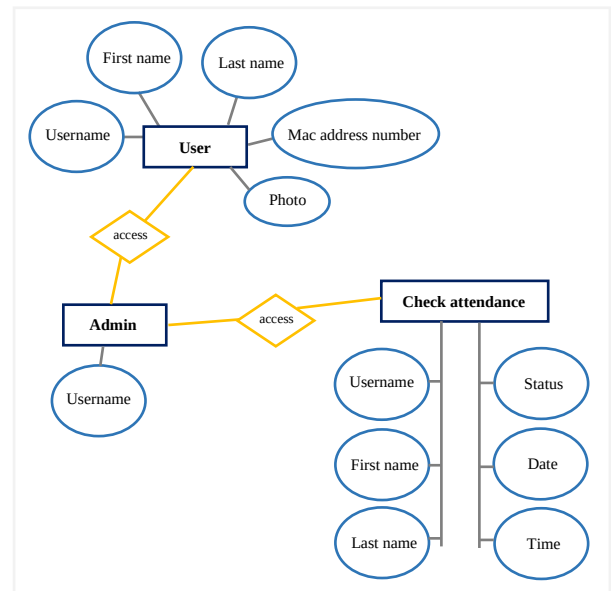
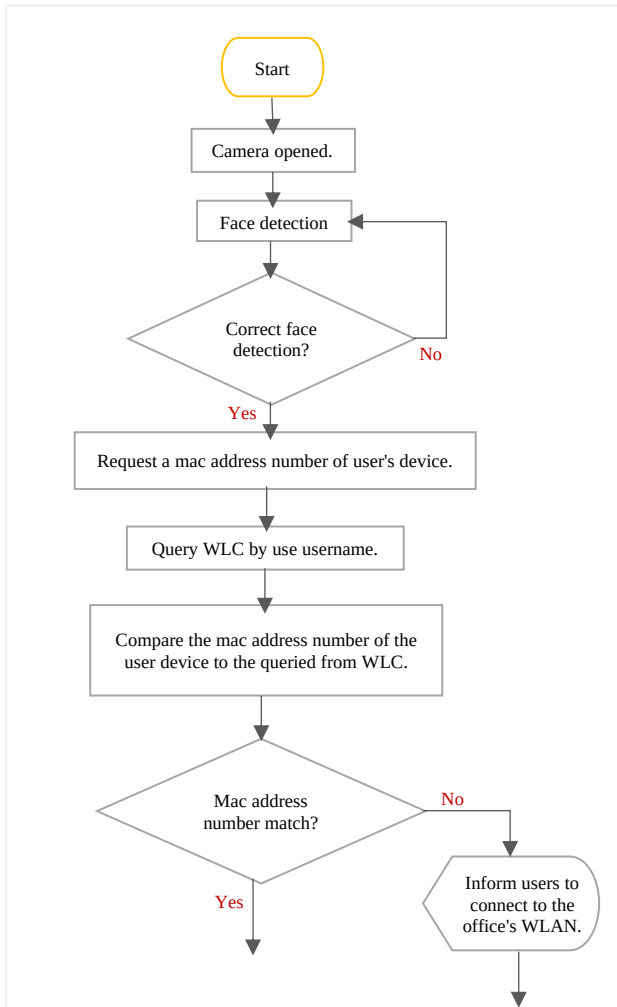


Fig. 2. Entity relationship diagram of database.



Fig. 3. Example of a user image stored in the database.

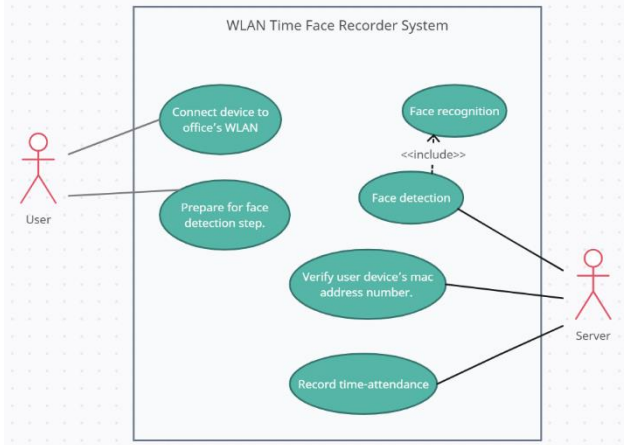


Fig. 4. Use case diagram.

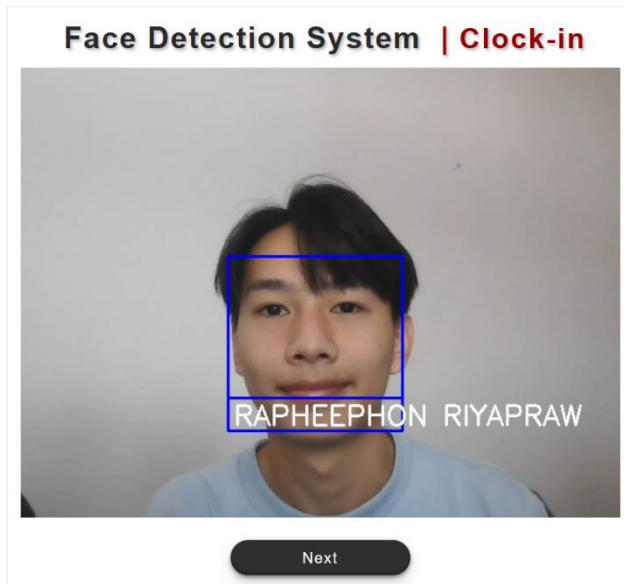


Fig. 5. Face detection system.

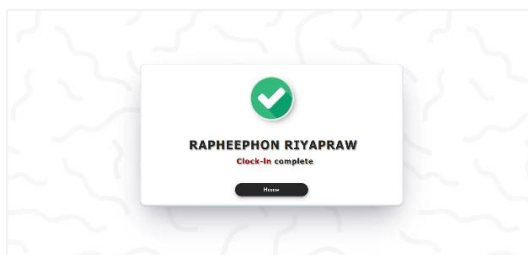


Fig. 6. Complete results page.



Fig. 7. Incomplete results page.

```

C:\Users\Computer>netsh wlan show interfaces | find "Physical address"
Physical address       : 14:5a:fc:01:f5:03
    
```

Fig. 8. Code for requesting the user's mac address number.

```

WLC#sh wireless client username 6331501057
Number of Clients: 1

MAC Address      AP Name      Status      Type ID      Auth Protocol
-----
145a.fc01.f503   AP1          Run         WLAN 1      Yes  11ax(2.4)
    
```

Fig. 9. Code to query WLC from username

```

WLC#sh wireless client usern
Number of Clients: 1

MAC Address      AP Name
-----
145a.fc01.f503   AP1
    
```

Fig. 10: user's mac address number from query WLC

Username	First name	Last name	Mac Address 1
6331501057	Rapheephon	Riyapraw	145a.fc01.f503

Fig. 11. user's mac address number from database

Username	First name	Last name	Status	Date	Time
6331501057	Rapheephon	Riyapraw	clock in	07/22/23	07:55:34

Fig. 12. Example of a time attendance table

II. PERFORMANCE EVALUATION

The WLAN time face attendance system is designed to streamline the time attendance process by enabling users to use their own devices instead of relying on a central machine. The main goal is to reduce waiting times in the check-in queue. The system has been evaluated through system testing to verify its performance and the results it produces.

A. Results and evaluation of the performance

To evaluate the performance and results, the system was put to the test with a group of 30 computer engineering students from Mae Fah Luang University. The relevant data for each student, including their username, first name, last name, MAC address, and picture, was stored in the system's database.

The results of the system testing showed that the MAC address lookup using the username, performed through

querying in the Wireless LAN Controller (WLC), produced correct results, as depicted in Fig. 13. and 14.

As for the test results, we have the results of processing time (s) and accuracy (%) as shown in Fig. 15. We have the results of processing time (s) versus accuracy (%) as shown in Fig. 16. The computer used for testing the system has the following device specifications: Processor: Intel(R) Core(TM) i7-10700 CPU @ 2.90GHz, Installed RAM 16.0 GB.

```
WLC#sh wireless client username 6331501057
Number of Clients: 1

MAC Address      AP Name
-----
145a.fc01.f503   AP1
```

Fig. 13: user's mac address number from query WLC

Username	First name	Last name	Mac Address 1
6331501057	Rapheephon	Riyapraw	145a.fc01.f503

Fig. 14: user's mac address number from query database

Accuracy (%)			Processing time (s)		
#1-10	#11-20	#21-30	#1-10	#11-20	#21-30
93.45	96.79	85.29	4.08	4.13	4.10
95.33	96.90	95.18	4.31	4.24	4.35
96.34	92.96	91.02	4.35	4.09	4.11
96.63	96.12	93.24	4.07	4.39	4.12
96.03	94.62	94.00	4.31	4.10	4.09
96.87	96.67	89.67	4.11	4.20	3.90
96.98	96.78	94.91	4.07	4.16	4.08
95.92	96.86	95.22	4.28	4.08	4.08
91.33	93.62	95.06	4.24	4.10	4.10
96.99	96.61	97.01	4.35	4.14	4.33

Fig. 15: The performance table.

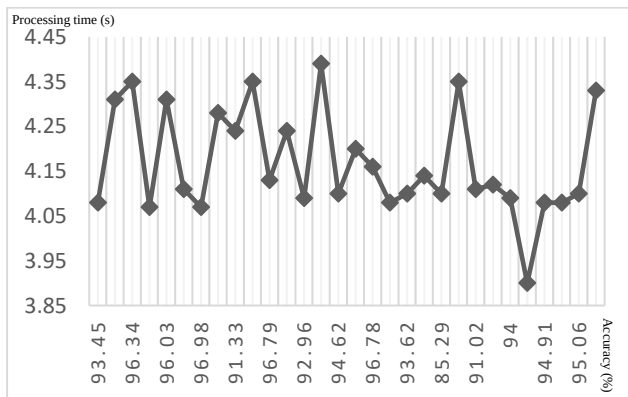


Fig. 16: The performance graph.

B. Realtime system precessing

Additionally, the system's face detection functionality accurately recognized and matched the user's face, as demonstrated in Fig. 17. and 18.

Based on these test results, it can be concluded that the proposed WLAN time face attendance system is efficient and effective. The accurate MAC address lookup and correct user face detection provide clear evidence that the system performs well and generates complete and accurate results.

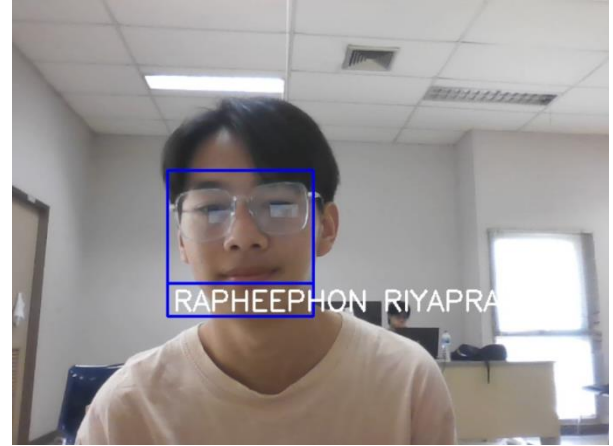


Fig. 17: The face detection system works correctly.



Fig. 18: Example of user image stored in the database.

C. System limitation

This limitation is due to WLC's ability to import username fields in some rows, which sometimes affects the system's functionality as Fig. 19. The reason is that user information is sent from another WLC, causing some information to be sent incomplete. When the system checks the user's WLAN office connection and the user has successfully connected, WLC does not record the username information, even though it records other information in this connection, such as the mac address number. Therefore, when the system queries this username, The result is that no devices are currently connected as Fig. 20.

AP Name	SSID	WLAN ID	Client Type	State	Protocol	User Name	Device Type
E3B-FL3-AP0208	MFUconnect	6	WLAN	Run	11ax(5)		Microsoft-Workstation
F2-FL4-AP0703	MFUconnect	6	WLAN	Run	11ax(5)		Linux-Workstation
BS-FL2-AP1392	MFUconnect	6	WLAN	Run	11n(2.4)		Android-Samsung
L1-FL2-AP0865	MFUconnect	6	WLAN	Run	11ax(5)		iPhone 11
AV-FL4-AP0064	MFUconnect	6	WLAN	Run	11ax(5)		IPad Air (4th Gen)
AV-FL3-AP0061	MFUconnect	6	WLAN	Run	11ax(5)		iPhone SE (2nd Gen)
Sak2-FL4-AP1548	MFUconnect	6	WLAN	Run	11ax(5)		iPhone 13
IP-FL2-AP1876	MFUconnect	6	WLAN	Run	11ax(5)		iPhone SE (3rd gen)
E1-FL2-AP0107	MFUconnect	6	WLAN	Run	11ac	4821	Microsoft-Workstation
E2-FL1-AP0150	MFUconnect	6	WLAN	Run	11ac	4821	Microsoft-Workstation
E3A-FL4-AP0194	MFUconnect	6	WLAN	Run	11ac	5121	Microsoft-Workstation
H0S-FL2-AP1808	MFUconnect	6	WLAN	Run	11n(2.4)	5221	Microsoft-Workstation

Fig. 19: WLAN connection information from WLC.

```
WLC#show wireless client username 6331501067
Number of Clients: 0
```

Fig. 20: The system queries username with incomplete information sent.

III. DISCUSSION and CONCLUSION

The integration of WLAN into the time attendance system has resulted in notable improvements in terms of ease of use and system reliability. With users being able to use their own devices for attendance, the process becomes more convenient and efficient, eliminating the need for central machines and reducing wait times in the check-in queue. Moreover, the WLAN connection verification adds an extra layer of authentication to the system, enhancing its reliability. By verifying the user's connection to the company's WLAN, the system can ensure that the individual is physically present within the company's premises when checking in, thereby providing an additional level of accuracy, and preventing attendance fraud.

We hope that The Development of WLAN Time Face Recorder System will be a pilot that can be developed and put into practice to increase convenience, and increase the reliability of the name-check-in system for users and reduce the cost of maintaining central equipment for the company.

ACKNOWLEDGEMENT

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